EXPERIMENT 1

ive/exp2.m

```
clc;
clear;
close all;

%% ============================================================
% CHARACTERISTICS STUDY OF SIDEBAND POWER AND FREQUENCIES IN AM
% USING MATLAB
% ============================================================

%% 1. INITIAL PARAMETERS

Fs = 1e6;              % Sampling frequency = 1 MHz
T  = 0.01;             % Simulation time = 10 ms

t = 0:1/Fs:T-1/Fs;

% Carrier and message parameters
fc = 100e3;            % Carrier frequency = 100 kHz
fm = 10e3;             % Message frequency = 10 kHz

Ac = 1;                % Carrier amplitude
Am = 1;                % Message amplitude

R = 50;                % Load resistance = 50 ohms

%% 2. MODULATION INDEX

mu = Am/Ac;

fprintf('\n');
fprintf('=================================================\n');
fprintf('       AM SIDEBAND POWER AND FREQUENCY STUDY\n');
fprintf('=================================================\n');
```

```
%% 2. MODULATION INDEX

mu = Am/Ac;

fprintf('\n');
fprintf('=================================================\n');
fprintf('       AM SIDEBAND POWER AND FREQUENCY STUDY\n');
fprintf('=================================================\n');

fprintf('\nBasic Parameters:\n');
fprintf('Carrier Frequency       = %.2f kHz\n',fc/1000);
fprintf('Message Frequency       = %.2f kHz\n',fm/1000);
fprintf('Carrier Amplitude       = %.2f V\n',Ac);
fprintf('Message Amplitude       = %.2f V\n',Am);
fprintf('Load Resistance         = %.2f Ohm\n',R);
fprintf('Modulation Index        = %.2f\n',mu);

%% 3. MESSAGE SIGNAL

message = Am*cos(2*pi*fm*t);

%% 4. CARRIER SIGNAL

carrier = Ac*cos(2*pi*fc*t);

%% 5. AM SIGNAL

AM_signal = Ac*(1 + mu*cos(2*pi*fm*t)) ...
           .* cos(2*pi*fc*t);

%% ============================================================
% 6. TIME DOMAIN PLOTS
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```
% Message signal

figure;

plot(t*1000,message,'LineWidth',1.2);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Message Signal');

grid on;

xlim([0 1]);

% Carrier signal

figure;

plot(t*1000,carrier,'LineWidth',1);

xlabel('Time (ms)');
ylabel('Amplitude (V)');

title('Carrier Signal');

grid on;

xlim([0 0.2]);

% AM signal

figure;
```

```
fprintf('\n');
fprintf('------------------------------------------------------\n');
fprintf('          RESULTS FOR 100%% MODULATION\n');
fprintf('------------------------------------------------------\n');

fprintf('Carrier Power            = %.6f W\n',Pc);

fprintf('USB Power                = %.6f W\n',P_USB_100);

fprintf('LSB Power                = %.6f W\n',P_LSB_100);

fprintf('Total Sideband Power     = %.6f W\n',P_SB_100);

fprintf('Total AM Power           = %.6f W\n',P_total_100);

fprintf('Transmission Efficiency  = %.2f %%\n',efficiency_100);

fprintf('------------------------------------------------------\n');
```

```
%% ------------------------------------------------------------
% 26. FINAL CONCLUSION
% ------------------------------------------------------------

fprintf('\n');
fprintf('CONCLUSION:\n');

fprintf(['1. USB frequency = fc + fm.\n']);

fprintf(['2. LSB frequency = fc - fm.\n']);

fprintf(['3. AM bandwidth = 2fm.\n']);

fprintf(['4. Sideband power increases with mu^2.\n']);

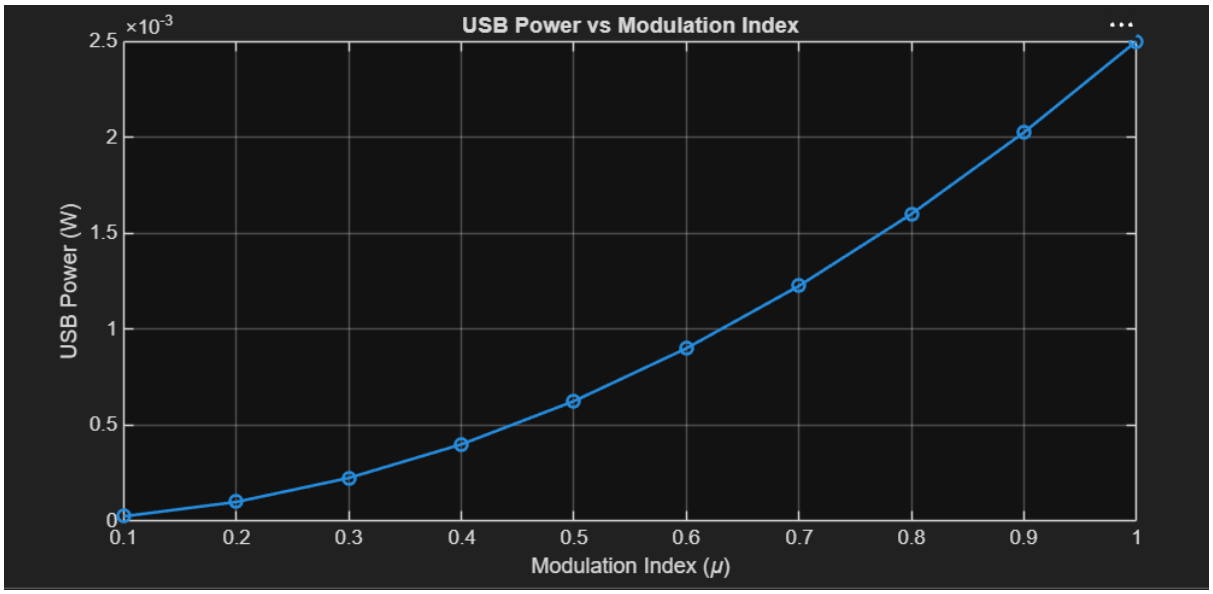
fprintf(['5. At 100%% modulation, total sideband power = Pc/2.\n']);

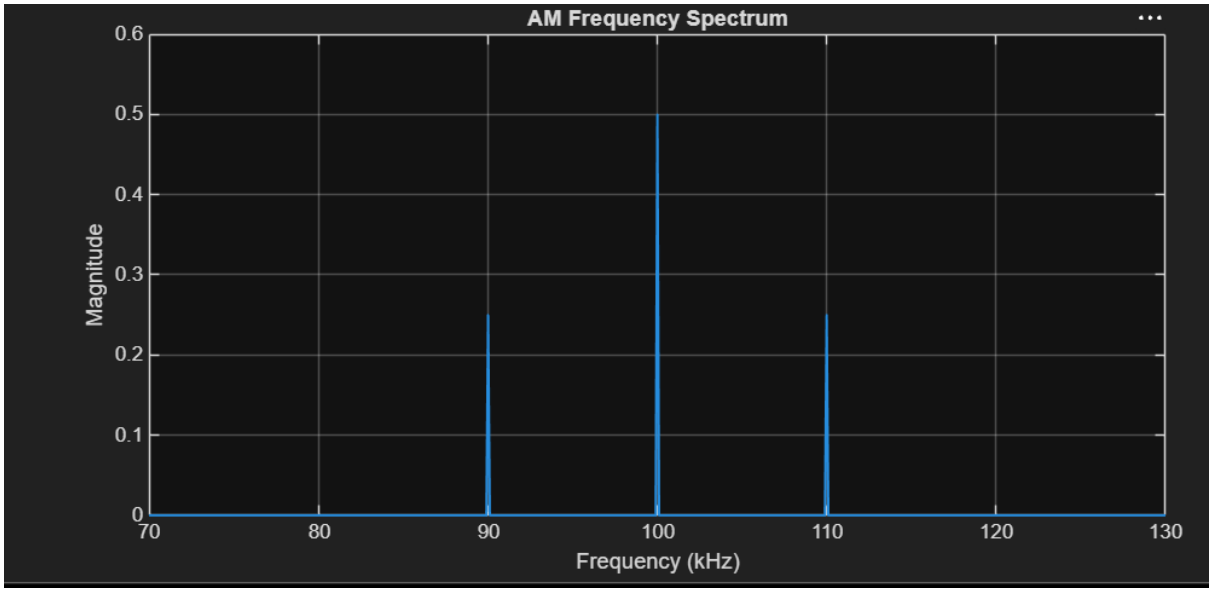
fprintf(['6. At 100%% modulation, total AM power = 1.5Pc.\n']);

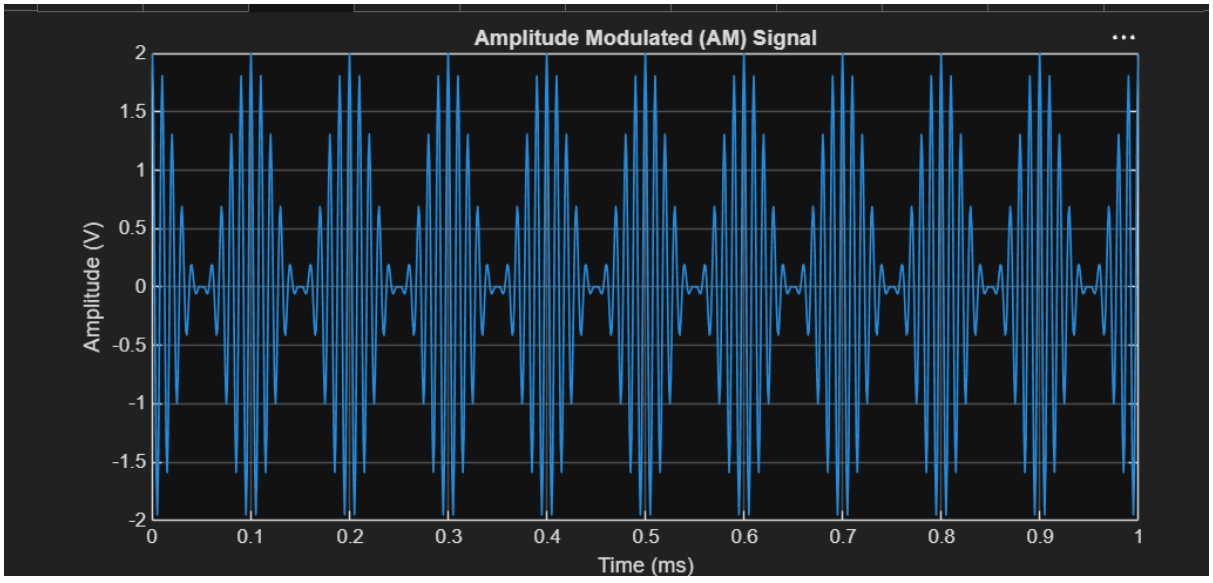
fprintf(['7. Maximum theoretical AM efficiency = 33.33%%.\n']);

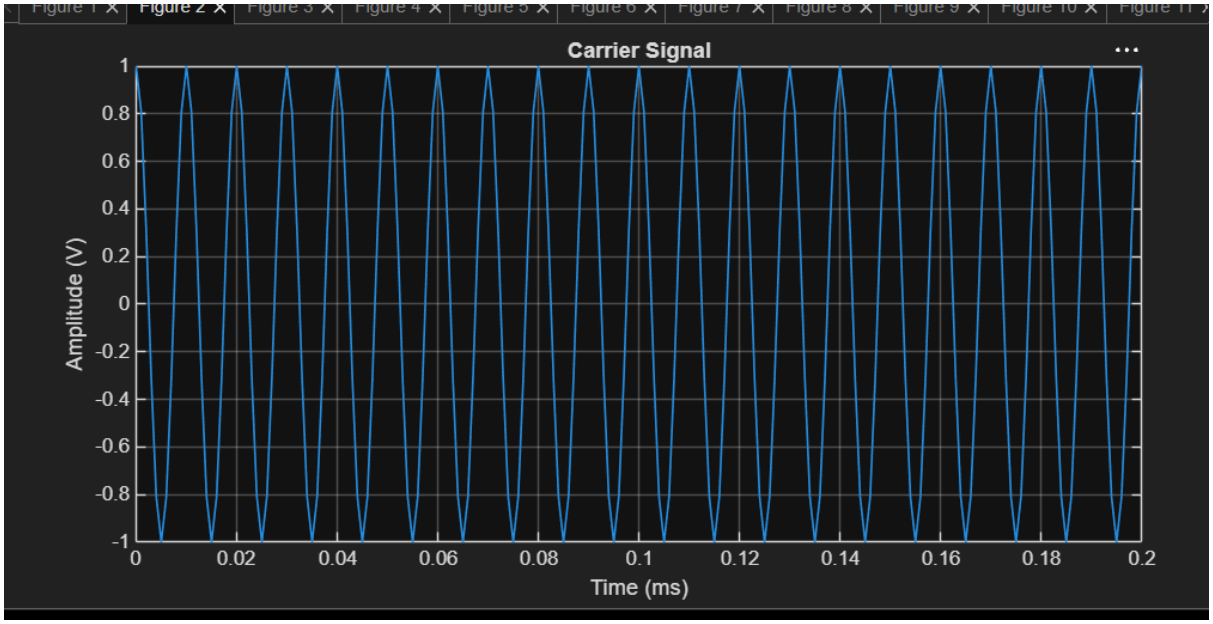
fprintf('\n------------------------------------------------------\n');
fprintf('             SIMULATION COMPLETED\n');
fprintf('------------------------------------------------------\n');
```

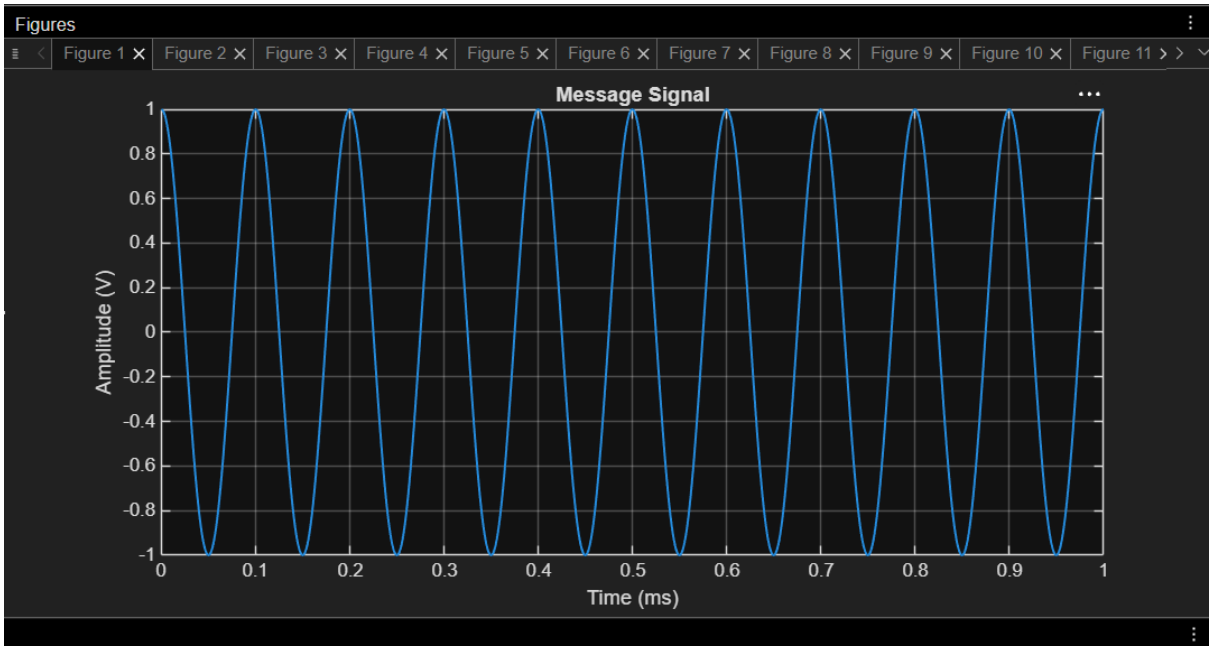
OUTPUT:

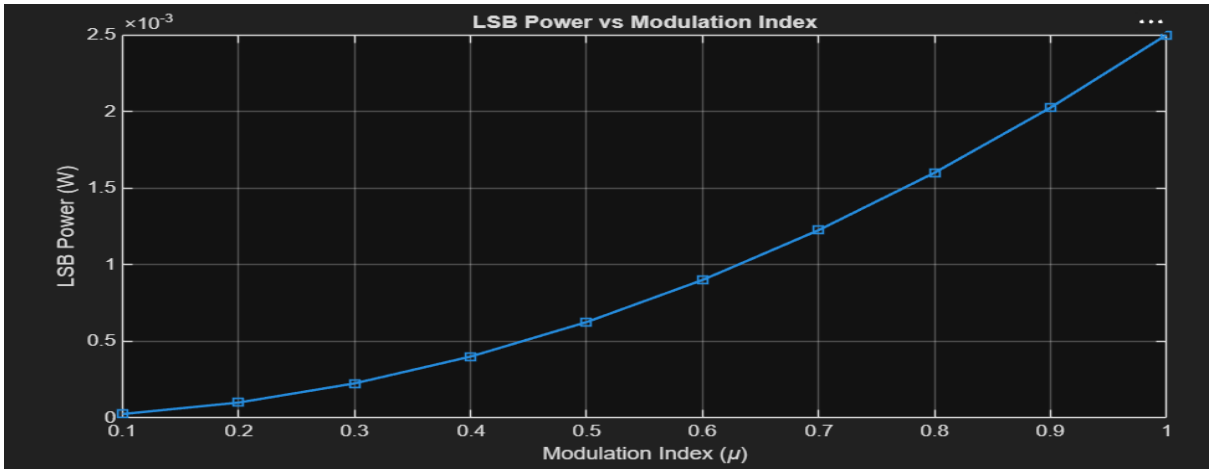

USB Power vs Modulation Index
×10^-3
USB Power (W)
Modulation Index (μ)
0
0.5
1
1.5
2
2.5
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1


AM Frequency Spectrum
Magnitude
Frequency (kHz)
0
0.1
0.2
0.3
0.4
0.5
0.6
70
80
90
100
110
120
130


Amplitude Modulated (AM) Signal
Amplitude (V)
Time (ms)
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1


Carrier Signal
Amplitude (V)
Time (ms)


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
Figure 7
Figure 8
Figure 9
Figure 10
Figure 11
Message Signal
Amplitude (V)
Time (ms)


LSB Power vs Modulation Index
×10-3
LSB Power (W)
Modulation Index (μ)